Generative AI Image Optimization & Model Behavior Analysis

Stable Diffusion | Prompt Engineering | LoRA Fine-Tuning

by Vinícius Henrique Conte

Objective

Evaluate limitations of a base generative model in producing culturally consistent urban environments and improve output quality using structured prompt engineering and LoRA fine-tuning.

Baseline Generation (Without LoRA)

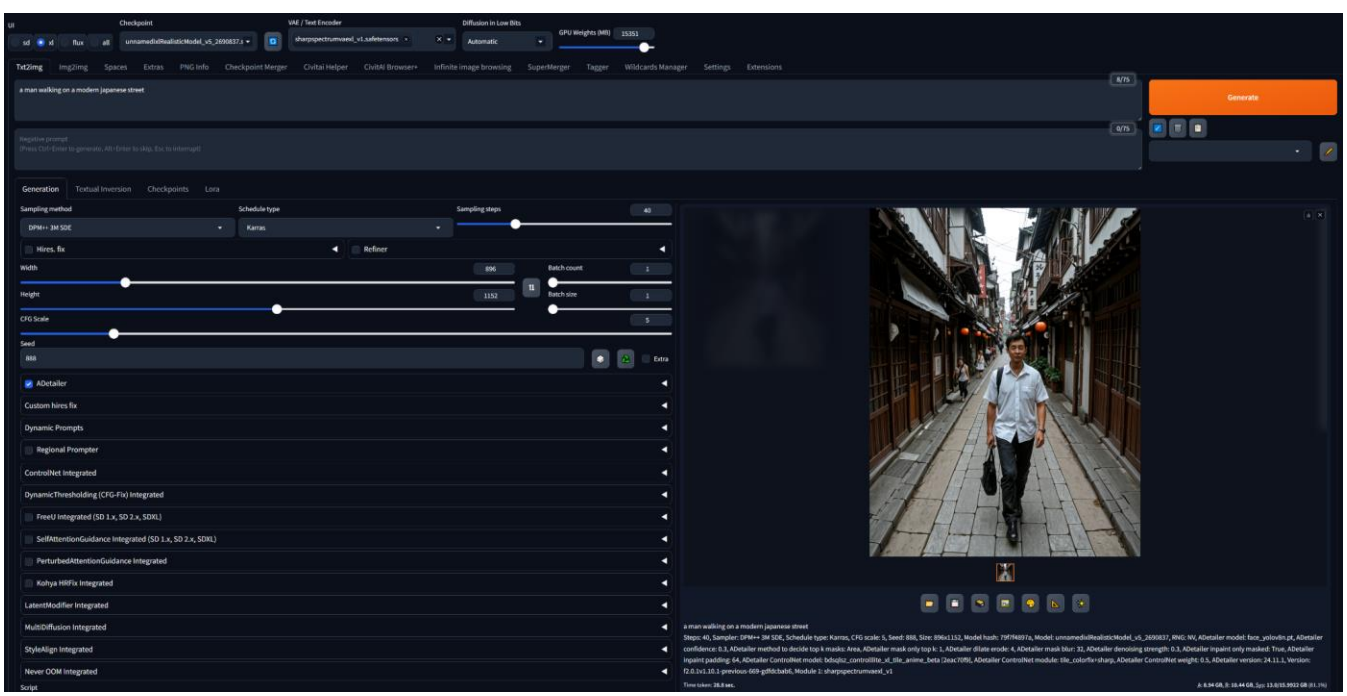
Prompt:

a man walking on a modern japanese street

Key Settings:

- Model: SDXL realistic checkpoint
- Sampler: DPM++ 3M SDE
- CFG Scale: 5
- Steps: 40

Generation Interface:



Output Result:

Observation: The base model produces a plausible image but lacks strong cultural consistency and detail in modern Japanese urban features.

Improved Generation (With LoRA & Structured Prompting)

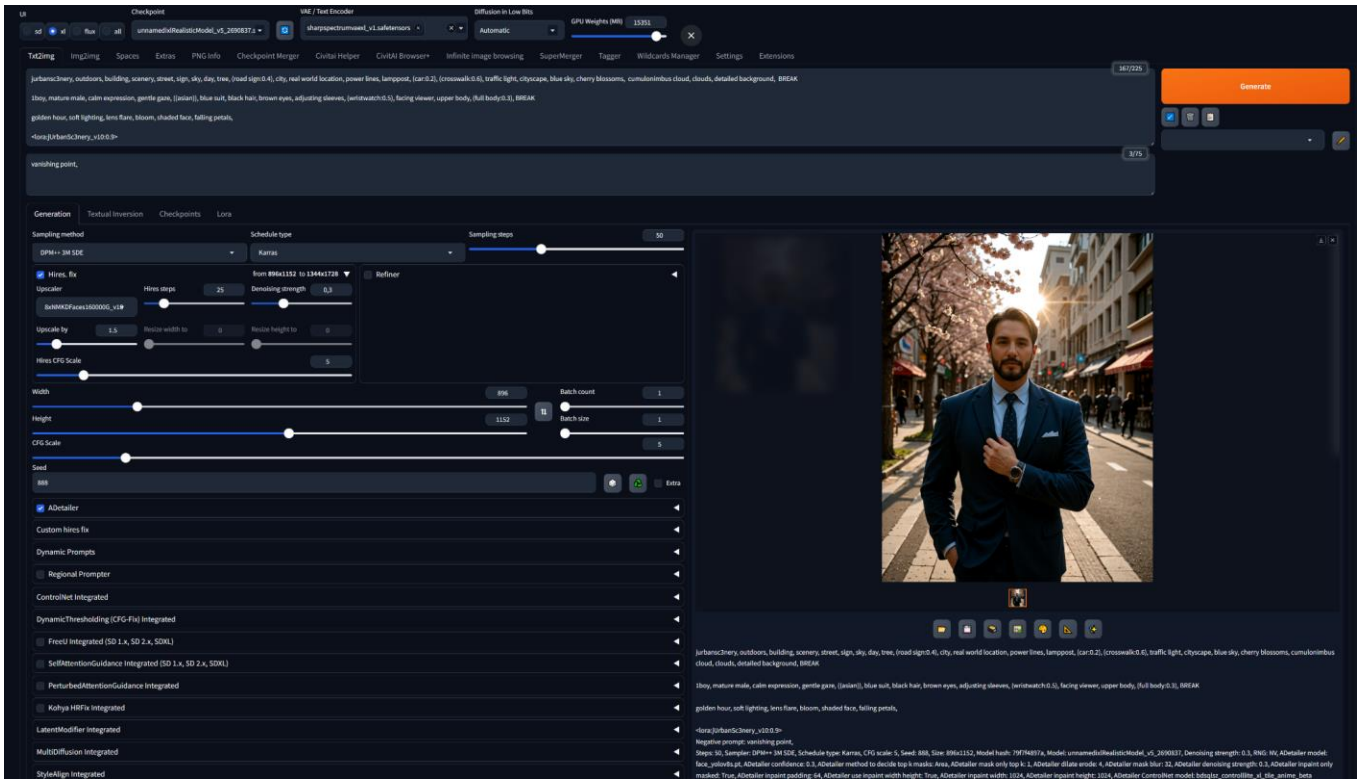
Enhancements Applied:

- Custom LoRA trained on ~40 curated images of modern Japanese streets
- Transition from natural language to structured Danbooru-style tags
- Prompt segmentation using BREAK (scene / subject / lighting)
- HiRes Fix for improved resolution
- ADetailer for refinement of facial and fine details

Structured Prompt Example:

japanese street, outdoors, buildings, narrow road, power lines, lanterns, detailed background BREAK 1boy, business attire, realistic face, upper body BREAK golden hour, soft lighting, depth of field

Generation Interface (with LoRA):



Output Result:

Result: Significant improvement in environmental coherence, lighting quality, and cultural accuracy. The LoRA enhances domain-specific representation that the base model lacked.

Technical Insight

This project highlights that prompt engineering alone may be insufficient when the model lacks domain representation. Combining structured prompting with LoRA fine-tuning enables better control, consistency, and realism.

Evaluation Criteria

- Visual consistency
- Environmental coherence
- Cultural accuracy
- Prompt adherence